

# Claude AI Learning Roadmap for AWS + DevOps Career Growth

This roadmap is designed for someone with your background:

- AWS Infrastructure experience
- Terraform knowledge
- GitHub and CI/CD exposure
- Limited coding background

Goal: Learn Claude AI in a practical way for cloud, DevOps, automation, troubleshooting, documentation, and career growth.

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## Phase 1: Understand Claude AI Basics (Week 1)

### Goal

Understand what Claude AI is and how it works.

### Topics to Learn

#### 1. What is Claude AI?

- Claude AI is an AI assistant created by Anthropic.
- Similar to ChatGPT.
- Used for:
  - Writing
  - Technical explanation
  - Coding help
  - DevOps troubleshooting
  - Automation ideas
  - Documentation

#### 2. Claude AI Versions

Learn the difference between:

- Claude Sonnet
- Claude Opus
- Claude Haiku

### 3. Where Claude AI is Used

- DevOps
- Cloud Engineering
- Automation
- Terraform support
- Architecture explanation
- Documentation generation
- Debugging

### 4. Claude AI Interface

Learn:

- Chat window
- Upload files
- Paste code
- Continue conversations
- Long context handling

### Practice Tasks

- Ask Claude to explain AWS services.
- Ask Claude to explain Terraform basics.
- Ask Claude to explain CI/CD.
- Ask Claude to explain Kubernetes.

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## Phase 2: Learn Prompt Engineering (Week 2-3)

### Goal

Learn how to ask Claude AI correctly.

Prompt engineering means giving proper instructions to AI.

### Topics to Learn

#### 1. Simple Prompt

Example:

"Explain AWS VPC in simple language"

## 2. Detailed Prompt

Example:

"Explain AWS VPC for beginners with real-world example and diagram explanation"

## 3. Role-Based Prompt

Example:

"Act as AWS Architect and explain ALB and NLB differences"

## 4. Step-by-Step Prompt

Example:

"Give me step-by-step Terraform setup for EC2 deployment"

## 5. Troubleshooting Prompt

Example:

"My EKS pod cannot connect to RDS. Help me troubleshoot step by step."

## 6. Prompt Chaining

Ask follow-up questions:

- Why?
- How?
- Show example.
- Give real-world scenario.

## Practice Tasks

Try asking Claude:

- Explain Terraform modules.
  - Explain GitHub Actions.
  - Explain EKS networking.
  - Explain IAM role trust relationship.
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# Phase 3: Claude AI for AWS Learning (Week 4-5)

## Goal

Use Claude to improve AWS knowledge.

## Topics to Learn

### 1. AWS Architecture Explanation

Ask Claude:

- Explain VPC architecture.
- Explain multi-tier architecture.
- Explain landing zone.
- Explain disaster recovery.

### 2. AWS Troubleshooting

Use Claude to debug:

- Security group issue
- Route table issue
- NAT gateway issue
- ALB issue
- EC2 connectivity issue

### 3. AWS Design Questions

Ask Claude:

- Design highly available architecture.
- Design secure architecture.
- Design multi-region architecture.

### 4. AWS Interview Questions

Use Claude for:

- AWS scenario-based interview preparation
- Real-world troubleshooting questions

## Practice Tasks

Practice daily:

- Ask 5 AWS questions.
  - Learn 1 AWS architecture daily.
  - Ask Claude to compare AWS services.
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## Phase 4: Claude AI for Terraform (Week 6)

### Goal

Use Claude to become stronger in Terraform.

### Topics to Learn

#### 1. Terraform Code Generation

Ask Claude:

- Create VPC Terraform code.
- Create EC2 Terraform code.
- Create ALB Terraform code.
- Create Auto Scaling Terraform code.

#### 2. Terraform Debugging

Ask Claude:

- Why Terraform apply failed?
- Explain Terraform error.
- Explain Terraform state.

#### 3. Terraform Best Practices

Learn:

- Variables
- Outputs
- Modules
- Remote backend
- Workspaces
- Environment separation

## 4. Terraform Review

Paste Terraform code and ask:

- Improve this code.
- Optimize this module.
- Find issue in code.

## Practice Tasks

Build:

- VPC
  - EC2
  - ALB
  - Auto Scaling
  - Security Groups
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# Phase 5: Claude AI for GitHub Actions and CI/CD (Week 7)

## Goal

Use Claude for pipeline creation.

## Topics to Learn

### 1. YAML Pipeline Creation

Ask Claude:

- Create GitHub Actions workflow.
- Create CI pipeline.
- Create CD pipeline.

### 2. CI/CD Troubleshooting

Ask Claude:

- Why build failed?
- Why deployment failed?
- Why artifact not generated?

### 3. Automation

Learn:

- Auto deployment
- Trigger-based pipeline
- Environment deployment
- Secret management

### 4. Pipeline Security

Learn:

- Secrets management
- IAM role integration
- GitHub OIDC

## Practice Tasks

Build:

- EC2 deployment pipeline
- Docker build pipeline
- Terraform deployment pipeline

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## Phase 6: Claude AI for DevOps Automation (Week 8–9)

### Goal

Use Claude to automate tasks.

### Topics to Learn

#### 1. Shell Script Help

Ask Claude:

- Write Linux commands.
- Write Bash scripts.
- Explain Linux troubleshooting.

## 2. Python for DevOps

Use Claude to learn:

- Python basics
- AWS Boto3 basics
- Automation scripts

## 3. Monitoring

Ask Claude:

- Explain CloudWatch.
- Explain monitoring design.
- Explain logging architecture.

## 4. Documentation

Claude can create:

- SOP documents
- KT documents
- Deployment guide
- Runbooks

## Practice Tasks

Ask Claude to create:

- Deployment checklist
- Troubleshooting guide
- Infrastructure documentation

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# Phase 7: Claude AI Power User (Week 10–11)

## Goal

Use Claude professionally.



# Topics to Learn

## 1. Upload Files

Upload:

- Terraform code
- Logs
- AWS architecture
- YAML files

Ask Claude to:

- Analyze
- Debug
- Improve

## 2. Large Project Understanding

Ask Claude to:

- Understand full Terraform project
- Explain folder structure
- Explain dependencies

## 3. Claude for Career Growth

Use Claude for:

- Resume optimization
- Interview preparation
- LinkedIn summary
- Job description understanding

## 4. Daily Claude Usage

Use Claude every day for:

- Learning
  - Troubleshooting
  - Documentation
  - Architecture planning
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# Daily Practice Plan

## Daily 30–45 Minutes

### Monday

AWS architecture

### Tuesday

Terraform

### Wednesday

GitHub Actions

### Thursday

DevOps troubleshooting

### Friday

Python basics

### Saturday

Build small project

### Sunday

Revision

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## Best Claude AI Prompts For Your Career

### AWS

"Explain AWS VPC architecture with real-world example"

### Terraform

"Create Terraform code for VPC, EC2, and ALB"

## GitHub Actions

"Create GitHub Actions workflow for Terraform deployment"

## DevOps

"Explain CI/CD pipeline in simple language"

## Troubleshooting

"My EKS pod cannot connect to RDS. Give me step-by-step troubleshooting"

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# Final Career Advantage

After learning Claude AI, you can become stronger in:

- AWS Infrastructure
  - Terraform
  - GitHub Actions
  - DevOps Automation
  - Architecture Design
  - Troubleshooting
  - Documentation
  - AI-Assisted Engineering
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# 3-Month Outcome

After 3 months of consistent learning, you should be able to:

- Use Claude as daily DevOps assistant
- Generate Terraform quickly
- Troubleshoot AWS issues faster
- Build CI/CD pipelines faster
- Improve interview performance
- Increase productivity in cloud projects
- Become AI-assisted cloud engineer